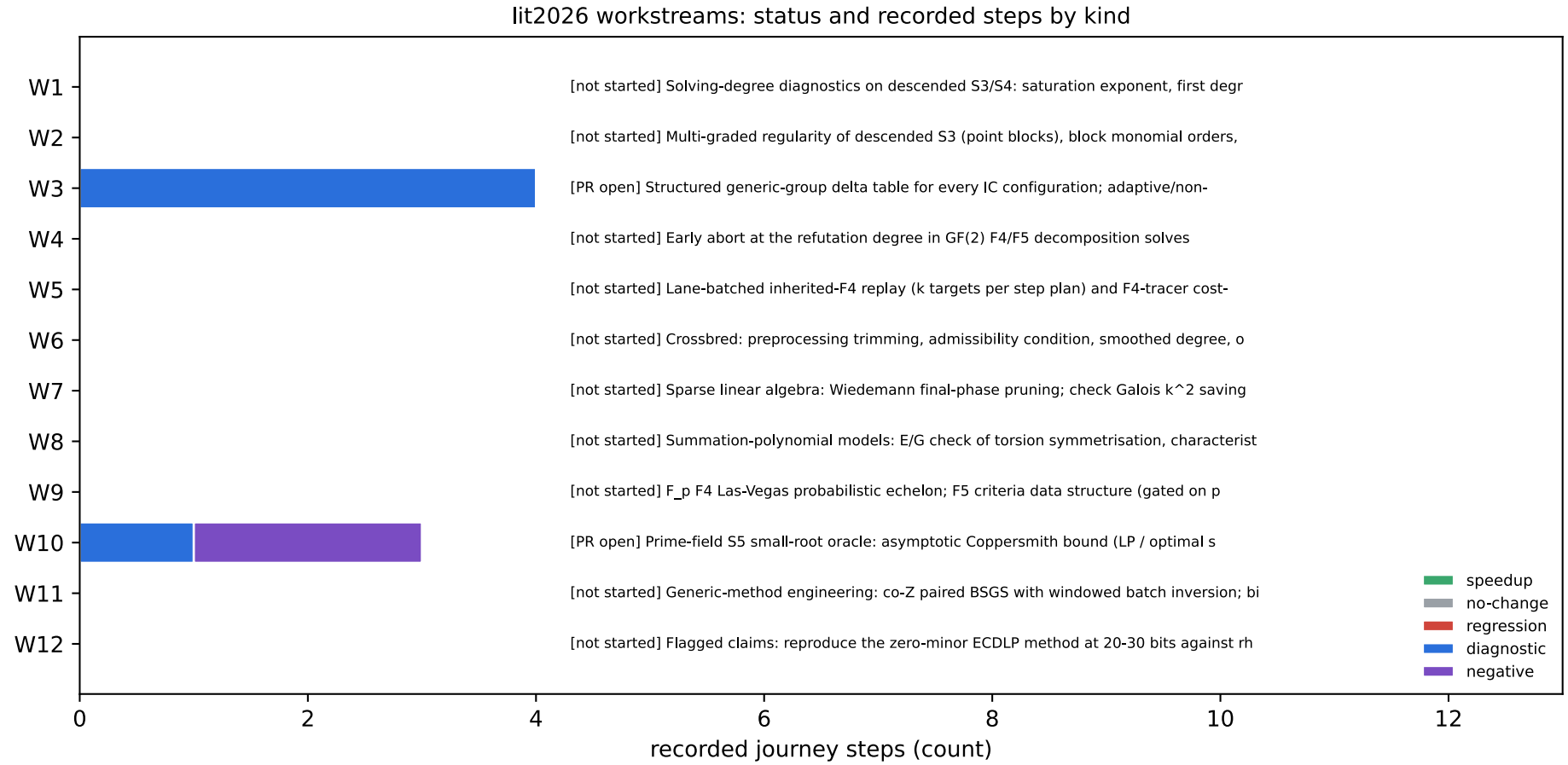
## lit2026 speedup program: journey and workstream board

Program baseline commit `4b6e09b78`. Every plotted ratio comes from a journey entry, measured in one interleaved session beside the baseline binary. Sources: `workstreams.json` and `journey/entries/`.




Source: research/lit2026_speedup_program/journey/entries/*.json

Figure 1: Measured program-baseline / candidate ratio per workload, by date.




Source: research/lit2026_speedup_program/workstreams.json and journey/entries/*.json

Figure 2: Workstream status and recorded steps by kind.

| ID | Kind | Workstream | Status / PRs |
|---|---|---|---|
| W1 | diagnostic | Solving-degree diagnostics on descended S3/S4: saturation exponent, first degree fall, generic leading-monomial divergence, algebraic-randomness Hilbert test<br>ePrint 2024/528, arXiv 2410.23211, arXiv 2505.10246, ePrint 2024/1924, arXiv 2609.14052 | not started |
| W2 | diagnostic+speedup | Multi-graded regularity of descended S3 (point blocks), block monomial orders, variable-order variance<br>ePrint 2025/2060, ePrint 2025/793, arXiv 2602.02972, arXiv 2509.14754 | not started |
| W3 | audit | Structured generic-group delta table for every IC configuration; adaptive/non-adaptive precompute classification; multi-target slope<br>ePrint 2026/384, arXiv 2505.00894, ePrint 2026/2253 | PR open aburan28/crypto#1683 |
| W4 | speedup | Early abort at the refutation degree in GF(2) F4/F5 decomposition solves<br>arXiv 2603.24890 | not started |

| ID | Kind | Workstream | Status / PRs |
|---|---|---|---|
| W5 | speedup | Lane-batched inherited-F4 replay (k targets per step plan) and F4-tracer cost-model check<br>arXiv 2607.06372, arXiv 2603.16378 | not started |
| W6 | speedup | Crossbred: preprocessing trimming, admissibility condition, smoothed degree, oracle multistep; Hamming-ball / decremental Boolean exhaustive phase<br>ePrint 2026/360, ePrint 2024/758, ePrint 2026/408, arXiv 2506.09950, ePrint 2026/040, ePrint 2026/1704, ePrint 2025/1788 | not started |
| W7 | speedup | Sparse linear algebra: Wiedemann final-phase pruning; check Galois k^2 saving in Koblitz linear algebra<br>ePrint 2026/1787, ePrint 2026/560 | not started |
| W8 | diagnostic | Summation-polynomial models: E/G check of torsion symmetrisation, characteristic-uniform model S3, Frobenius auxiliary variable, Lucas-identity splitting<br>ePrint 2026/1299, arXiv 2608.01675, ePrint 2026/1927, arXiv 2609.18866 | not started |
| W9 | speedup | F_p F4 Las-Vegas probabilistic echelon; F5 criteria data structure (gated on profile share); tracer zero-reduction predictor<br>arXiv 2609.14433, arXiv 2601.05026, arXiv 2505.23696 | not started |
| W10 | feasibility | Prime-field S5 small-root oracle: asymptotic Coppersmith bound (LP / optimal shifts) against delta = 1/5<br>ePrint 2024/1577, ePrint 2026/1027, ePrint 2026/423 | PR open aburan28/crypto#1691 |
| W11 | speedup | Generic-method engineering: co-Z paired BSGS with windowed batch inversion; binary x-only isogeny evaluation; k=2 biextension pairing; DP hit-rate statistics; external challenge ladder<br>ePrint 2026/2102, ePrint 2026/704, ePrint 2025/1652, ePrint 2026/1823, arXiv 2508.14011 | not started |
| W12 | audit | Flagged claims: reproduce the zero-minor ECDLP method at 20-30 bits against rho; record ePrint 2025/015 as unimplementable from its abstract<br>arXiv 2607.09814, ePrint 2025/015 | not started |